

Hybrid Quantum-Classical Architecture for Solving Large-Scale QUBO Problems

A Divide et Impera Approach

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Da un secolo, oltre.

QUBO problems

QUBO formulation

$$\min_{x \in \{0,1\}^n} f(x) = x^T Q x = \sum_{i=1}^n \sum_{j=1}^n Q_{i,j} x_i x_j \quad (1)$$

where:

- $x = (x_1, x_2, \dots, x_n)$ is a vector of n binary decision variables;
- Q is an $n \times n$ square matrix of real constants detailing the problem's costs and interactions.
- $x_i^2 = x_i$ for any $x_i \in \{0, 1\}$. Thus, any linear term of the objective function can be mapped directly onto the main diagonal of the Q matrix ($Q_{i,i}$), while the off-diagonal elements ($Q_{i,j}$ for $i \neq j$) encode the quadratic interactions, or "penalties", between pairs of variables.

Physical Formulation

QUBO to Ising Hamiltonian Mapping (1/2)

The system's energy is governed by the Ising Hamiltonian. While it may appear intimidating, it breaks down into two distinct parts that map natively to the QUBO framework:

$$\hat{\mathcal{H}}(t) = \hbar \sum_{i=1}^N \left(\frac{\Omega(t)}{2} \hat{\sigma}_i^x - \delta(t) \hat{n}_i \right) + \sum_{i < j} \frac{C_6}{|r_i - r_j|^6} \hat{n}_i \hat{n}_j \quad (2)$$

1. Single-Atom Driving (Linear Terms):

- $\hbar$ is the reduced Planck constant, scaling angular frequencies to energy;
- N is the total number of atoms (qubits) in the register;
- $\Omega(t)$ is the time-dependent Rabi frequency (laser amplitude);
- $\hat{\sigma}_i^x$ is the Pauli-X operator driving the transition between the ground state $|0\rangle$ and the Rydberg state $|1\rangle$;
- $\delta(t)$ is the detuning parameter of the laser frequency.

Physical Formulation

QUBO to Ising Hamiltonian Mapping (2/2)

$$\hat{\mathcal{H}}(t) = \hbar \sum_{i=1}^N \left(\frac{\Omega(t)}{2} \hat{\sigma}_i^x - \delta(t) \hat{n}_i \right) + \sum_{i < j} \frac{C_6}{|r_i - r_j|^6} \hat{n}_i \hat{n}_j \quad (3)$$

2. Pairwise Interactions (Quadratic Penalties):

- $\hat{n}_i$ is the occupation number operator ($|1\rangle_i \langle 1|_i$), evaluating to 1 if atom i is in the Rydberg state, and 0 otherwise;
- r_i is the spatial position of atom i ;
- C_6 is the Van der Waals interaction coefficient, a platform-dependent constant.

Neutral Atom Quantum Computing

Components

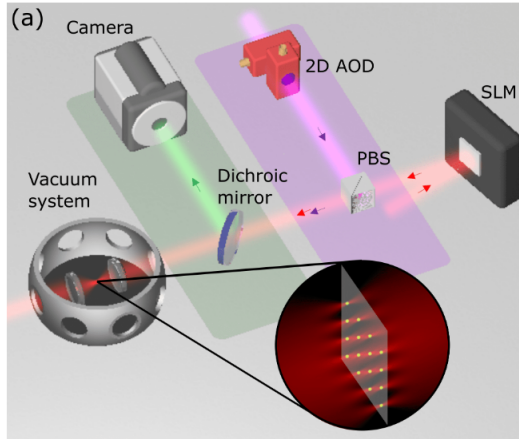


Figure: Image adapted from: L. Henriet et al., "Quantum computing with neutral atoms," *Quantum*, vol. 4 (2020)

Neutral Atom Quantum Computing

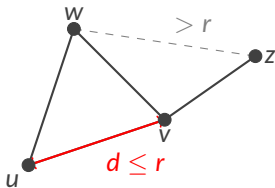
Context

- **The Paradigm:** Neutral atoms trapped in optical tweezers act as qubits, offering long coherence times and dynamic 2D/3D topologies.
- **Analog Mode & QUBO:** The physical dynamics (Ising Hamiltonian) perfectly map to Quadratic Unconstrained Binary Optimization (QUBO) problems.
- **The Hardware Limit:** Exciting atoms to the Rydberg state induces the *Rydberg blockade*. This naturally enforces constraints, making it ideal for Maximum Independent Set (MIS) problems on Unit Disk Graphs (UDG).

Embeddable problems: Unit Disk Graphs

UDG Problems formalization and visualization

(a) Distance Model



(b) Intersection Model

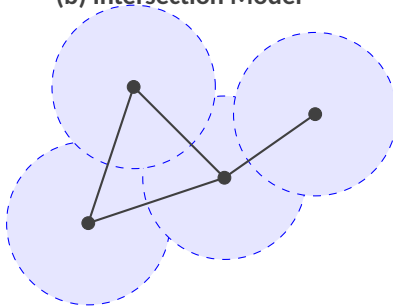


Figure: The equivalent geometric representations of a Unit Disk Graph (UDG). **(a)** The Distance Model connects nodes u and v if their Euclidean distance is $d \leq r$. **(b)** The Intersection Model places a disk of radius $r/2$ centered on each node; edges are formed exclusively where disks overlap.

The Embedding Problem

Hardware Constraints (Pasqal *Jade*)

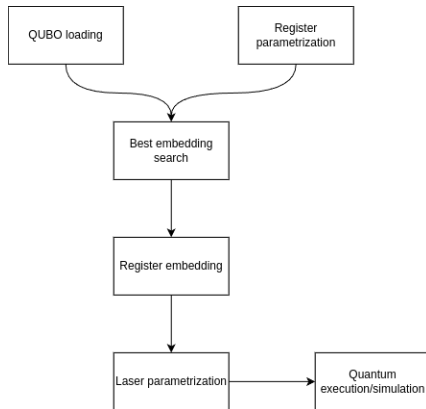
To execute a QUBO problem, logical variables must be mapped onto physical 2D coordinates. This spatial embedding is an NP-Hard problem itself.

Hardware Constraints (Pasqal *Jade*):

- **Maximum Radius ($R_{\max}$):** Register confined within $50.0 \mu\text{m}$.
- **Minimum Distance ($d_{\min}$):** Atoms cannot be closer than $4.0 \mu\text{m}$ (diffraction limit).
- **Global Laser Parameterization:** Uniform detuning and Rabi frequency for all atoms.

Baseline Architecture

Pipeline Overview



Embedding search

Genetic Algorithms & Simulated Annealing

We implemented a baseline hybrid pipeline to test monolithic embeddings:

- **Genetic Algorithms (GA):** Evolving 2D coordinates. Works for small instances, but crossover operations are physically destructive on dense spatial layouts.
- **Simulated Annealing (SA):** A single-trajectory thermodynamic approach enhanced with a **GRASP initialization** and zero-temperature **Quenching**.

The Evaluation Metric: A custom Improved Topological MAE fitness function that handles maximum physical potential clipping and bounds crossover.

Baseline Results Roadmap

How to Read the Experimental Data

The following slides evaluate the scaling limits of monolithic embeddings (GA vs. SA). Each test is presented in a standardized dual-panel layout:

Left Panel: Spatial Embedding

- The physical 2D coordinate layout found.
- Light shaded areas represent the physical atomic radius, visually confirming compliance with the minimum distance constraint (d_{min}).

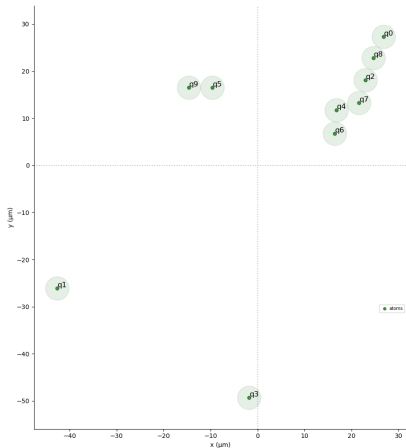
Right Panel: Analog Quantum Readout

- The probability distribution of the actual bitstrings measured after the adiabatic emulation.
- **Green Bars:** The *true mathematical ground states*, verified via exact classical benchmarking.
- **Blue Bars:** Sub-optimal measurements, representing excited states or false local minima.

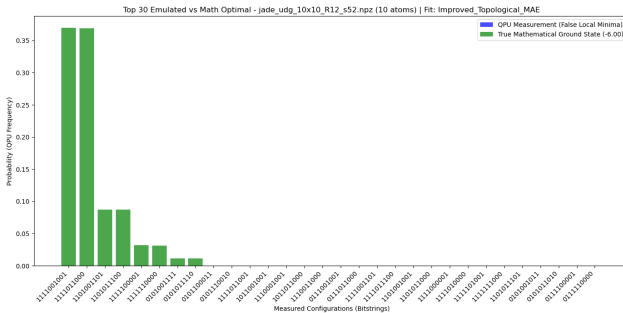
Baseline Results

GA on a 10x10 Instance

Spatial Embedding



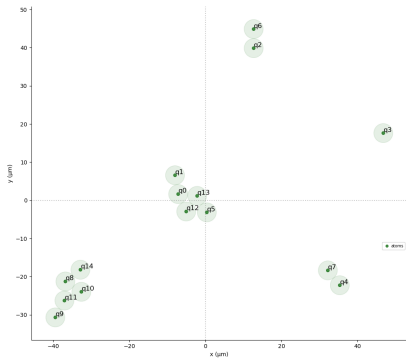
Quantum Readout



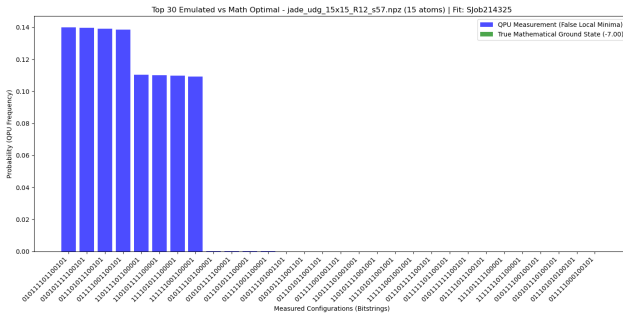
Baseline Results

Scaling limits: GA on a 15x15 Instance

Spatial Embedding



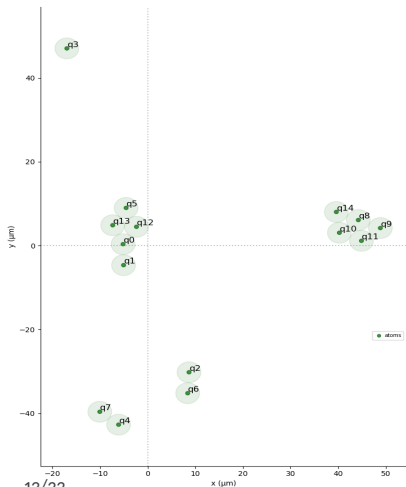
Quantum Readout



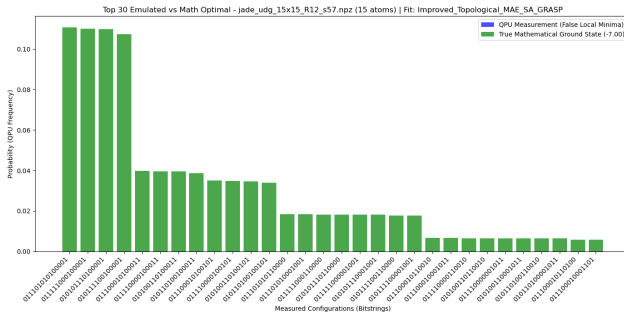
Baseline Results

SA on a 15x15 Instance

Spatial Embedding



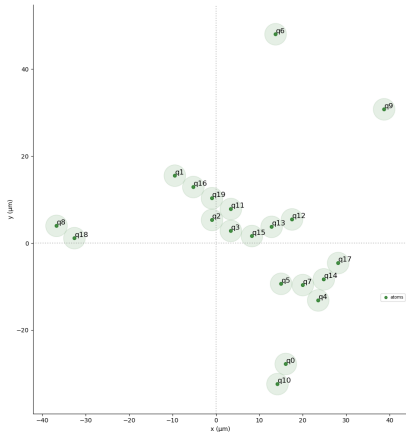
Quantum Readout



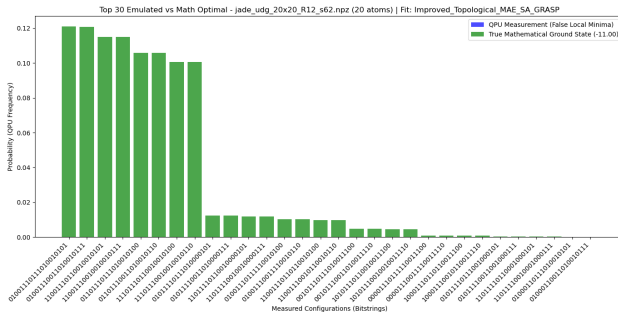
Baseline Results

SA on a 20x20 Instance

Spatial Embedding

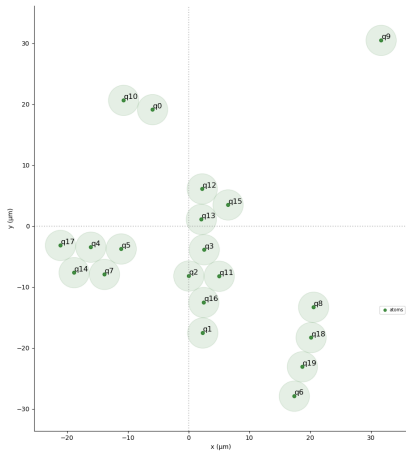


Quantum Readout

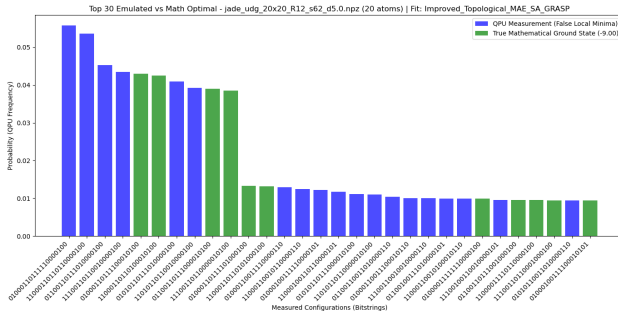


SA on a more difficult 20x20 Instance

Spatial Embedding



Quantum Readout



The Scalability Wall

The Operational Breaking Point

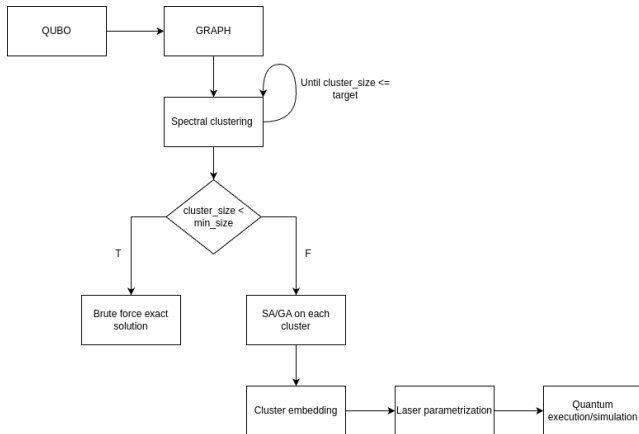
The Operational Breaking Point

Despite hardware resilience in quantum emulation, classical monolithic embedding heuristics face the *spatial curse of dimensionality*.

- GA pipeline fails on 15×15 instances.
- SA pipeline hits a computational wall on dense 20×20 matrices, leading to degraded spatial fitness and noisy quantum readouts.
- **Conclusion:** With better classical hardware we can surely scale more, but the embedding of larger problems benefits from a paradigm shift.

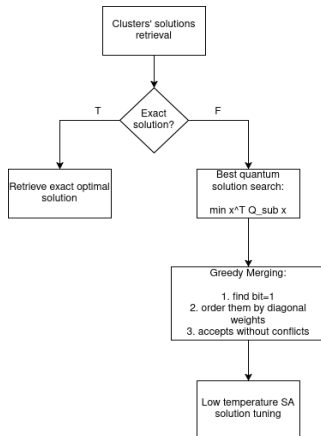
The *Divide et Impera* Architecture

Partitioning Phase



The *Divide et Impera* Architecture

Merging Phase



Experimental Results

Pushing the frontier: 100×100 Instance

Target: Pushing the operational frontier on a highly dense UDG QUBO matrix.

1. Topological Profile & Routing

Metric	Value
Variables (Nodes)	100
Graph Density	4.8%
Max Node Degree	10
Total Sub-clusters	10
Cluster Sizes	From 2 to 15 nodes
Execution Routing	1 Exact, 9 QPU (SA)

2. Energy Convergence

Reconstruction Phase	Global Energy
Baseline Greedy (No filter)	-29.0
Filtered Greedy (Noise mitigation)	-35.0
Final Low-Temp SA Tuning	-37.0*

* *Perfectly matches the true global minimum independently approximated via D-Wave benchmark.*

Experimental Results

Extreme saturation: More difficult 100 × 100 Instance

Target: Pushing the structural limits of the architecture under a big instance pressure.

1. Topological Profile & Partitioning

Metric	Value
Variables (Nodes)	100
Graph Density	6.6%
Max Node Degree	13 (Extreme Saturation)
Total Sub-clusters	9
Cluster Sizes	From 5 to 17 nodes
Execution Routing	o Exact , 9 QPU (SA)

2. Strategy & Energy Convergence

Reconstruction Phase	Global Energy
Filtered Greedy (Noise mitigation)	−26.0
Final Low-Temp SA Tuning	−30.0*

* *Perfectly matches the true global minimum independently approximated via D-Wave benchmark.*

Conclusions

Take-Home Messages

- **Current Hardware Viability:** Neutral Atom platforms have proven highly promising, capable of tackling combinatorial optimization problems beyond toy-dimension instances.
- **The Necessity of Partitioning:** Monolithic embeddings hit strict physical ceilings. The *Divide et Impera* architecture bypasses continuous spatial constraints by transforming intractable global configurations into manageable micro-tasks.
- **Future-Proof Modularity:** The distributed approach is inherently modular. As hardware capabilities expand, the architecture scales seamlessly without requiring structural redesigns.

Thank you for your attention!

Q&A

Future Outlook

Moving Beyond Analog Emulation

- **Direct Physical Deployment:** Overcoming the current reliance on HPC emulators by submitting tasks directly to a physical Neutral Atom QPU, evaluating empirical resilience against raw physical noise.
- **Advanced Laser Pulse Optimization:** Exploring Optimal Control Theory (OCT) or Analog QAOA to dynamically shape the Rabi frequency and detuning profiles, minimizing diabatic transitions and boosting ground-state probability.
- **Alternative Problem Formulations:** Expanding the orchestrator to automatically translate higher-level combinatorial problems (e.g., Integer Programming) into platform-compliant UDG topologies.

Future Outlook

Algorithmic Enhancements

- **Alternative Clustering & Orchestration:** Investigating hybridization with Multi-Agent Systems (MAS) or Reinforcement Learning agents for problem clustering, local evaluation, and pipeline orchestration.
- **Quantum-Native Boundary Resolution:** "Freezing" internal atoms that reached an optimal state and isolating strictly the conflicting *border graph*. Resubmitting this border to the QPU fully delegates conflict resolution to the quantum hardware.

Appendix: GA Parametrization

Hyperparameters and Results for the 10×10 Instance

Execution Metrics: The classical preprocessing required 1408.8 seconds, achieving a spatial fitness score of 0.0583.

Category	Parameter	Value
Evolution	Population Size	300
	Number of Generations	1500
Operators	Mutation Probability	[0.10, 0.02]
	Random Mutation Window	$[-1.0, 1.0] \mu m$
	Elitism	15 individuals
Mechanisms	Selection Method	K_Tournament (Size = 4)
	Fitness Evaluation	Improved Topological MAE

Appendix: GA Parametrization

Hyperparameters and Results for the 15×15 Instance

Execution Metrics: The classical preprocessing required 2032.72 seconds, achieving a severely degraded spatial fitness score of 0.0075.

Category	Parameter	Value
Evolution	Population Size	400
	Number of Generations	3000
Operators	Mutation Probability	[0.10, 0.01]
	Random Mutation Window	$[-1.0, 1.0] \mu m$
	Elitism	20 individuals
Mechanisms	Selection Method	Tournament (Size = 5)
	Mating Pool Size	10 parents

Appendix: SA Parametrization

Hyperparameters and Results for the 15×15 Instance

Execution Metrics: The thermodynamic engine required 2429.1 seconds, yielding a spatial fitness score of 0.04305.

Category	Parameter	Value
Thermodynamic Schedule	Initial Temperature (T_{init})	50000.0
	Minimum Temperature (T_{min})	0.1
	Cooling Rate	0.99
Exploration	MCMC Steps per Temperature	1000
	Independent Restarts	10

Appendix: SA Parametrization

Hyperparameters and Results for the 20×20 Instance

Execution Metrics: A brute-force thermodynamic search taking 7669.73 seconds, with a resulting spatial fitness score of 0.0102.

Category	Parameter	Value
Thermodynamic Schedule	Initial Temperature (T_{init})	50000.0
	Minimum Temperature (T_{min})	0.1
	Cooling Rate	0.99
Exploration	MCMC Steps per Temperature	2000
	Independent Restarts	15

Appendix: SA Parametrization

Hyperparameters and Results for the Denser 20 × 20 Instance

Execution Metrics: The conservative execution required 2125.04 seconds, yielding a degraded spatial fitness score of 0.0039.

Category	Parameter	Value
Thermodynamic Schedule	Initial Temperature (T_{init})	50.0
	Minimum Temperature (T_{min})	0.01
	Cooling Rate	0.98
Exploration	MCMC Steps per Temperature	800
	Independent Restarts	15

Appendix: The Van der Waals Interaction

Physical Constraints and the Rydberg Blockade

The interaction between pairs of neutral atoms in the Rydberg state is governed by the Van der Waals potential for dipole-dipole interactions:

$$V_{ij} = \frac{C_6}{r_{ij}^6}$$

Physical and Mathematical Implications:

- **Variables:** C_6 is a constant dependent on the specific atomic species and excitation level, while r_{ij} is the spatial Euclidean distance between atoms i and j .
- **The Rydberg Blockade:** As the distance r_{ij} decreases, the repulsive energy V_{ij} scales massively. This interaction strictly prevents two spatially close atoms from being simultaneously excited to the Rydberg state.
- **QUBO Mapping:** This physical penalty natively acts as the quadratic off-diagonal terms ($Q_{i,j}$) of the combinatorial matrix. It heavily suppresses configurations that violate the logical boundaries of the target graph.

Appendix: Fitness Function Formulation

Improved Topological MAE (1/2)

The custom fitness function evaluates the geometric quality of a candidate layout and its structural isomorphism to the target QUBO matrix.

1. Physical Potential Clipping:

As $r_{ij} \rightarrow 0$, the physical potential explodes. To prevent numerical instability, the effective potential $\tilde{V}_{ij}$ is bounded using a saturation threshold $n = 5$:

$$\tilde{V}_{ij} = \min \left(\frac{C_6}{r_{ij}^6}, n \cdot Q_{max} \right)$$

2. Topological Base Error (E_{base}):

Edges are partitioned into active ($> \epsilon$) and inactive ($\leq \epsilon$) sets using a tolerance $\epsilon = 10^{-5}$. The error penalizes deviations on active bonds and unwanted crosstalk on inactive ones:

$$E_{base} = \sum_{(i,j) \in E_{active}} |\tilde{V}_{ij} - Q_{ij}| + \lambda \sum_{(i,j) \in E_{inactive}} \tilde{V}_{ij}$$

The amplification multiplier $\lambda = 2.0$ forces the algorithm to move unconnected nodes apart.

Appendix: Fitness Function Formulation

Improved Topological MAE (2/2)

3. Geometric Hardware Penalty (P):

The layout must strictly respect the optical tweezers' diffraction limits ($d_{min} = 4.0 \mu\text{m}$) and the global laser's maximum radius ($R_{max} = 50.0 \mu\text{m}$):

$$P = K \left(\sum_{i < j} \max(0, d_{min} - r_{ij}) + \sum_i \max(0, ||r_i|| - R_{max}) \right)$$

A massive penalty factor $K = 10^5$ immediately drops physically unfeasible candidates from the evolutionary pool.

4. Final Fitness Value (F):

Because GAs natively maximize a score, the accumulated errors and penalties are inverted:

$$F = \frac{1}{E_{base} + P + \delta}$$

A minimal constant $\delta = 10^{-6}$ is introduced to prevent zero-division crashes.

Appendix: Simulated Annealing Engine

Energy Formulation and Spatial Perturbations

SA operates as a probabilistic, single-trajectory metaheuristic. It optimizes the continuous spatial layout by iteratively minimizing an internal energy $E(S)$:

$$E(S) = E_{base} + P$$

1. Geometric Neighborhood Moves ($S \rightarrow S'$):

- **Jitter (60%)**: Micro-displacement of a single atom. The maximum radius scales proportionally with the current temperature.
- **Swap (20%)**: The coordinates of two distinct atoms are perfectly swapped, resolving topological entanglements.
- **Jump (20%)**: An atom is randomly replaced anywhere within the permissible R_{max} boundary for global exploration.

2. Thermodynamic Evolution:

Transitions to a worse configuration ($\Delta E > 0$) are accepted following the Boltzmann distribution to escape local minima:

$$P(S \rightarrow S') = e^{-\frac{\Delta E}{T}}$$

Appendix: Spectral Clustering

Graph Translation and Laplacian Matrix

Randomly splitting the QUBO matrix would sever crucial quadratic interactions. Instead, we translate it into an Affinity Matrix A :

$$A_{ij} = |Q_{ij}|(1 - \delta_{ij})$$

The Kronecker delta (δ_{ij}) forces $A_{ii} = 0$, ignoring linear biases to focus exclusively on the pure topological interconnectivity.

Next, we compute the Degree Matrix D and the unnormalized Graph Laplacian L :

$$D_{ii} = \sum_j A_{ij}$$

$$L = D - A$$

The eigenvector associated with the second smallest eigenvalue of L (the *Fiedler vector*) provides the optimal continuous relaxation for the discrete graph partitioning problem.

Appendix: Spectral Clustering

Algorithm and Recursive Size-Capping

To enforce the hardware's maximum qubit capacity, we dynamically calculate the number of target clusters k :

$$k = \max(2, \text{round}(N/\text{target_capacity}))$$

Algorithm: Spectral Graph Clustering

1. Create a sparsified similarity graph G (represented by matrix A).
2. Compute the graph Laplacian for G , denoted as L .
3. Create a matrix V consisting of the first k eigenvectors of L .
4. Apply K-means clustering on the rows of V to obtain the k discrete clusters.

Recursive execution: If any resulting partition still exceeds the maximum hardware capacity, the algorithm recursively calls itself exclusively on that overgrown sub-graph.

Appendix: SA Parametrization

Three-Tiered Configuration for the 100×100 Instance

Execution Metrics: The deterministic greedy merging yielded -29.0 , filtering improved it to -35.0 , and the final SA tuning successfully reached the true global minimum of -37.0 .

Parameter	Large (13-15 nodes)	Standard (9-11 nodes)	Small (7-8 nodes)
Restarts	5	5	5
Initial Temperature (T_{init})	1000.0	100.0	80.0
Minimum Temperature (T_{min})	0.01	0.01	0.10
Cooling Rate	0.99	0.98	0.95
MCMC Steps per Temperature	1200	1000	300

Appendix: SA Parametrization

Four-Tiered Configuration for the Denser 100×100 Instance

Execution Metrics: Greedy merging yielded -26.0 . SA post-processing elegantly refined the boundary intersections, successfully reaching the optimal -30.0 .

Parameter	Extreme (17 nodes)	Large (14-16 nodes)	Standard (9-12 nodes)	Small (5-7 nodes)
Restarts	5	5	5	5
Initial Temperature (T_{init})	2000.0	1000.0	100.0	80.0
Minimum Temperature (T_{min})	0.01	0.01	0.01	0.10
Cooling Rate	0.995	0.990	0.980	0.950
MCMC Steps per Temperature	2000	1200	1000	300

Appendix: Global Merging Refinement

Low-Temperature SA Post-Processing

While the deterministic greedy merging efficiently resolves boundary overlaps, it is prone to settling into sub-optimal configurations at the cluster intersections.

To refine the global state, the pipeline applies a final localized perturbation using Simulated Annealing:

- **Hot-Start Initialization:** Unlike standard SA, this phase does not begin from a random state. It directly injects the greedy-merged bitstring as the starting configuration.
- **Restricted Thermal Range:** By bounding the inverse temperature parameter (β) to a narrow, high-value range, the engine performs strictly localized, low-temperature micro-adjustments.
- **Structure Preservation:** This thermodynamic fine-tuning effectively squeezes out the remaining topological errors without destroying the highly optimized global structure assembled in the preceding steps.